

Question Paper Preview

Notations :

- 1.Options shown in **green** color and with  icon are correct.
- 2.Options shown in **red** color and with  icon are incorrect.

Change Font Color :	No
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MLT

Section Id :	640653124645
Number of Questions :	15
Section Marks :	50
Maximum Instruction Time :	0

Question Number : 1 Question Id : 6406531862037 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING TECHNIQUES (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536032623.  YES

6406536032624.  NO

Question Number : 2 Question Id : 6406531862038 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

For the polynomial kernel

$$K(f, g) = (f^T g + 1)^2,$$

the explicit feature map for the vector $f = (1, 2, -1)$ is 10-dimensional. Which of the following options is the correct transformed feature vector $\phi(f)$?

Options :

6406536032625. ✖ $(1, 1, 2, -1, 1, 4, 1, 2, -1, -2)$

6406536032626. ✔ $(1, \sqrt{2}, 2\sqrt{2}, -\sqrt{2}, 1, 4, 1, 2\sqrt{2}, -\sqrt{2}, -2\sqrt{2})$

6406536032627. ✖ $(1, 1, 4, 1, 2\sqrt{2}, -\sqrt{2}, -2\sqrt{2}, 0, 0, 0)$

6406536032628. ✖ $(1, \sqrt{2}, \sqrt{2}, \sqrt{2}, 1, 1, 1, 1, 1, 1)$

Question Number : 3 Question Id : 6406531862039 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

A centered dataset in $\mathbb{R}^4$ is projected onto its first two principal components. The first and the second principal components are respectively.

$$v_1 = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad v_2 = \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix},$$

Then, which of the following vectors could be a valid third principal component?

Options :

6406536032629. ✓ $\frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$

6406536032630. ✓ $\frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$

6406536032631. ✗ $\frac{1}{4} \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix}$

6406536032632. ✗

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Question Id : 6406531862040 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (4 to 5)

Question Label : Comprehension

Consider the following points on the real line:

$$x_1 = 1, x_2 = 4, x_3 = 7, x_4 = 8.$$

We run k -means algorithm with $k = 2$. The initial cluster centers are $\mu_1^0 = x_1$ and $\mu_2^0 = x_2$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 4 Question Id : 6406531862041 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

What will be the cluster means after iteration 1?

Options :

6406536032633. ✖ $\mu_1^{(1)} = 2, \mu_2^{(2)} = 6$

6406536032634. ✖ $\mu_1^{(1)} = 1, \mu_2^{(2)} = 7.5$

6406536032635. ✖ $\mu_1^{(1)} = 4, \mu_2^{(2)} = 7$

6406536032636. ✔ $\mu_1^{(1)} = 1, \mu_2^{(2)} = 6.33$

Question Number : 5 Question Id : 6406531862042 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Find the value of the objective function

$$F(z_1, z_2, z_3, z_4) = \sum_{i=1}^4 ||x_i - \mu_{z_i}||^2$$

after convergence. Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

8.63 to 8.68

Question Number : 6 Question Id : 6406531862043 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Suppose we have a $\text{Uniform}(1, 10)$ prior for the parameter θ . Then, the value of the maximum likelihood estimate (MLE) is equal to the value of the maximum a posteriori (MAP) estimate.

Options :

6406536032638. ✓ True

6406536032639. ✗ False

Question Number : 7 Question Id : 6406531862044 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Consider the following training dataset for a binary classification problem:

$$\mathcal{D} = \{(u, 1), (2u, 0), (3u, 1), (4u, 0)\},$$

where $u \in \mathbb{R}^d$ is a fixed nonzero constant and labels belong to $\{0, 1\}$. Let w be the weight vector of a linear classifier. What condition should the weight vector satisfy for the 0-1 loss to be zero on this dataset?

Options :

6406536032640. ✗ $w^\top u < 0$

6406536032641. ✗ $w^\top u > 0$

6406536032642. ✗ $w^\top u = 0$

6406536032643. ✓ No such weight vector w exists.

Question Number : 8 Question Id : 6406531862045 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

In Bayesian linear regression, suppose the prior distribution over the weight vector is

$$w \sim \mathcal{N}(0, \alpha^{-1}I).$$

Which of the following optimization problems is equivalent to finding the Maximum A Posteriori (MAP) estimate of w ?

Options :

6406536032644. ✖ $\|y - Xw\|$

6406536032645. ✖ $\|y - Xw\|^2 + \alpha\|w\|$

6406536032646. ✔ $\|y - Xw\|^2 + \alpha\|w\|^2$

6406536032647. ✖ $\|y - Xw\|^2 - \alpha\|w\|^2$

Question Number : 9 Question Id : 6406531862046 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a naive Bayes classification problem with three binary features f_1, f_2, f_3 for a binary classification problem. You are given the following 8 training samples:

Sample	f_1	f_2	f_3	y
x_1	0	1	0	0
x_2	1	1	0	0
x_3	1	1	1	1
x_4	1	0	0	1
x_5	1	0	1	0
x_6	0	1	1	1
x_7	1	1	0	1
x_8	0	0	1	0

Compute the Naive Bayes parameter estimate $\hat{p}(f_2 = 1 \mid y = 1)$ (without Laplace smoothing). Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.75

Question Number : 10 Question Id : 6406531862047 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Given a training set $\{(x_i, y_i)\}_{i=1}^n$, where $x_i \in \mathbb{R}^d$ and $y_i \in \{0, 1\}$, we want to estimate the parameter w for the logistic regression model. Following is the update rule for estimating w :

$$w^{t+1} = w^t + \eta \left[\sum_{i=1}^n x_i \left(y_i - \frac{1}{1 + e^{-w_t^T x_i}} \right) \right].$$

Select the correct options from the following:

Options :

6406536032649. ✓ If $y_i = 1$, and $\frac{1}{1 + e^{-w_t^T x_i}}$ is close to 1, then such points will not contribute much to the update of w .

6406536032650. ✗ If $y_i = 1$, and $\frac{1}{1 + e^{-w_t^T x_i}}$ is close to 0, then such points will not contribute much to the update of w .

6406536032651. ✗ If $y_i = 0$, and $\frac{1}{1 + e^{-w_t^T x_i}}$ is close to 1, then such points will not contribute much to the update of w .

6406536032652. ✓ If $y_i = 0$, and $\frac{1}{1 + e^{-w_t^T x_i}}$ is close to 0, then such points will not contribute much to the update of w .

Question Number : 11 Question Id : 6406531862048 Question Type : SA

Correct Marks : 4

Question Label : Short Answer Question

Consider a binary logistic regression model with the sigmoid function

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

Given a single training sample (x, y) where $x = \begin{bmatrix} 1.5 \\ -0.8 \end{bmatrix}$, $y = 1$, and current weight vector $w = \begin{bmatrix} 0.4 \\ 0.6 \end{bmatrix}$, compute the cross-entropy loss for this sample given by

$$L(w) = -[y \log_{10}(\hat{y}) + (1 - y) \log_{10}(1 - \hat{y})], \text{ where } \hat{y} = \sigma(w^T x).$$

Write your answer correct to three decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.273 to 0.279

Question Number : 12 Question Id : 6406531862049 Question Type : SA

Correct Marks : 4

Question Label : Short Answer Question

Consider an SVM classifier with separating hyperplane defined by $w^T x = 0$, where the weight vector is $w^T = \begin{bmatrix} 2 & -3 & 1 \end{bmatrix}$. Compute the perpendicular distance from the point $x^T = \begin{bmatrix} 1 & 2 & -1 \end{bmatrix}$ to the hyperplane. Enter your answer correct to three decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1.333 to 1.339

Question Number : 13 Question Id : 6406531862050 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider two datasets $D^{(1)}$ and $D^{(2)}$ where
 $D^{(1)} = \{(x_1^{(1)}, y_1^{(1)}), \dots, (x_n^{(1)}, y_n^{(1)})\}$

and

$$D^{(2)} = \{(x_1^{(2)}, y_1^{(2)}), \dots, (x_m^{(2)}, y_m^{(2)})\}$$

such that $x_i^{(1)} \in \mathbb{R}^{d_1}$ and $x_i^{(2)} \in \mathbb{R}^{d_2}$, where $d_1 > d_2$ and $n > m$. The maximum number of mistakes made by the perceptron algorithm is higher on dataset $D^{(1)}$ than on dataset $D^{(2)}$.

Options :

6406536032655. ✖ True

6406536032656. ✔ False

Question Number : 14 Question Id : 6406531862051 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

In a random forest classifier with p total features, each split is chosen by evaluating a randomly selected subset of m features. Consider how changing m affects the behavior of the forest. Which of the following statements are correct?

Options :

6406536032657. ✖ Increasing m reduces the correlation between any two trees in the forest.

6406536032658. ✔ Decreasing m reduces the correlation between any two trees in the forest.

6406536032659. ✔ If m is too small, individual trees may underfit the data.

6406536032660. ✖ Increasing m always decreases the variance of the random forest model.

6406536032661. ✖ Decreasing m makes trees more similar to each other.

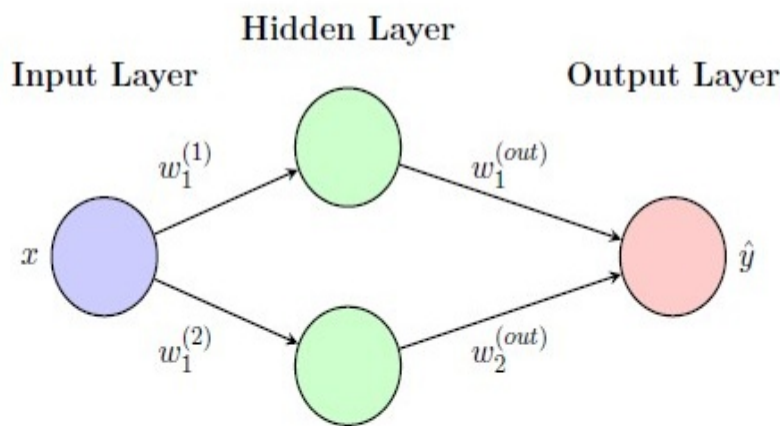
Question Id : 6406531862052 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (15 to 16)

Question Label : Comprehension

Consider the following neural network, which takes a single real-valued input x , passes it through two hidden ReLU units, and produces a scalar regression output. ReLU and the sigmoid activation is given by:

$$\text{ReLU}(x) = \max(0, x)$$
$$\sigma(x) = \frac{1}{1 + e^{-x}}.$$

The network structure is shown below.



Assume the following weights:

$$w_1^{(1)} = 0.2, \quad w_1^{(2)} = -0.4, \quad w_1^{(out)} = 1.0, \quad w_2^{(out)} = -0.5.$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 15 Question Id : 6406531862053 Question Type : SA

Correct Marks : 4

Question Label : Short Answer Question

For input $x = 0.5$, compute the the regression output $\hat{y}$. Enter the answer correct to one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.1

Question Number : 16 Question Id : 6406531862054 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Suppose the target value for this input is $y = 1.2$. Compute the squared loss $L = (\hat{y} - y)^2$. Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1.19 to 1.23

Question Id : 6406531862055 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (17 to 18)

Question Label : Comprehension

Consider the following one-dimensional dataset with binary labels $y \in \{+1, -1\}$.

Points	x	y
1	1	+1
2	2	+1
3	3	-1
4	4	-1
5	5	+1

AdaBoost uses decision stumps as weak learners. Assume the initial distribution of the dataset assigns equal weights to all data points, i.e., $D_0(i) = \frac{1}{5}$, for $i = 1, 2, 3, 4, 5$. At iteration $t = 1$, AdaBoost chooses the decision stump

$$h_1(x) = \begin{cases} +1, & x < 3, \\ -1, & x \geq 3. \end{cases}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 17 Question Id : 6406531862056 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which point will be misclassified by the first decision stump?

Options :

6406536032664. ✖ Points 4

6406536032665. ✔ Points 5

6406536032666. ✖ Points 3

6406536032667. ✖ Points 1

Question Number : 18 Question Id : 6406531862057 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

After AdaBoost updates the sample weights for iteration $t = 2$, which pair of points will have the smallest (and equal) weights while training the second stump h_2 ?

Options :

6406536032668. ✔ Points 1 and 2

6406536032669. ✔ Points 3 and 4

6406536032670. ✖ Points 3 and 5

6406536032671. ✖ Points 1 and 5